IAM (IDENTITY AND ACCESS MANAGEMENT)

IAM (Identity and Access Management) in AWS is a service that allows you to securely control access to AWS resources by managing users, groups, roles, and permissions, ensuring that only authorized entities can access specific services and actions in an AWS account.

- IAM is used to manage access to AWS resources securely.
- It allows creation of users, roles, and assignment of permissions.
- Permissions define what actions a user can perform on AWS services.
- IAM improves security by allowing only authorized access.

FACTORS OF IAM

[1] **Users** – Individual identities created to access AWS resources.

[2] **Groups** – Collection of users with similar permissions.

[3] **Roles** – Temporary permissions assigned to services or users.

[4] **Policies** – Rules that define what actions are allowed or denied.

IAM SECURITY PROCESSES

Authentication

- Verifies the identity of an IAM user.
- Done using username & password or access keys.
- MFA (Multi-Factor Authentication) can be enabled for extra security.

Authorization

- Determines what actions the authenticated user is allowed to perform.
- Controlled using IAM policies.
- Policies define allowed or denied access to AWS resources.

IAM USER IN AMAZON WEB SERVICES

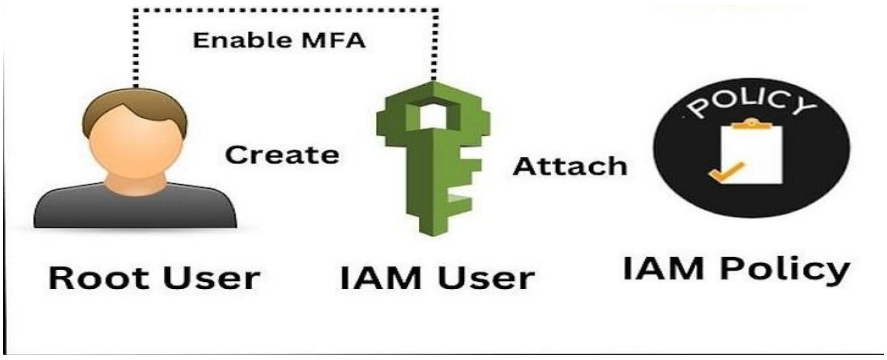
- An IAM user is an identity created to represent a person or application.
- It provides access to AWS services using a username and password or access keys.
- Permissions for an IAM user are assigned using policies.
- IAM users help manage secure and controlled access to AWS resources.

IAM GROUP IN AMAZON WEB SERVICES

- An **IAM group** is a collection of IAM users.
- It helps **manage permissions for multiple users at once**.
- Policies attached to a group are automatically applied to all users in that group.

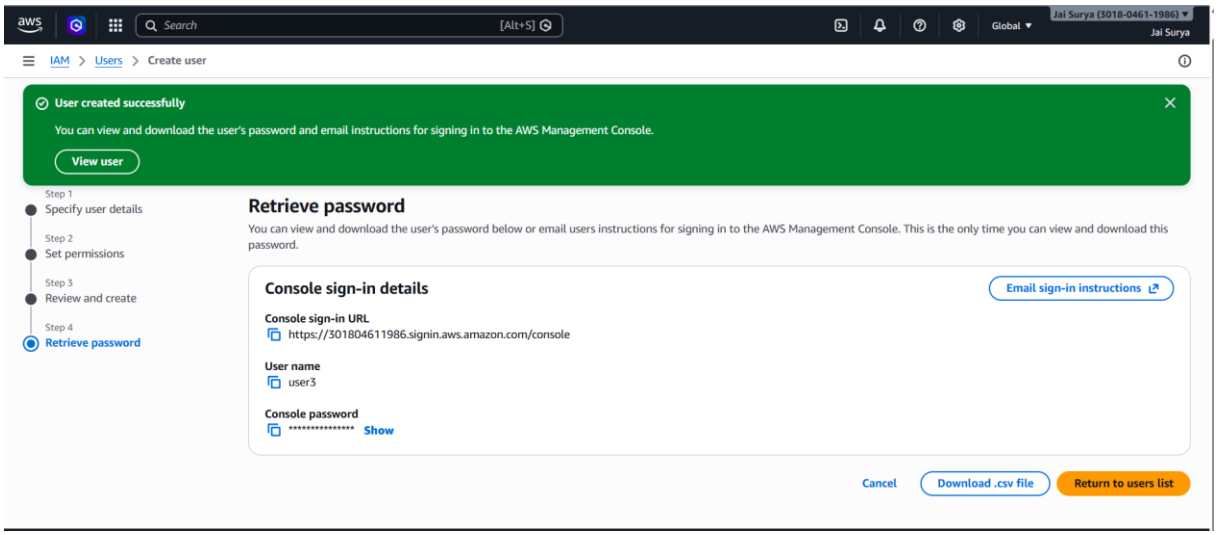
ROOT USER

- The root user is the account created when you first sign up for AWS.
- Root user can create IAM users and groups.
- Root user has full permissions to access all resources and services.
- Root user can assign permissions to IAM users and groups based on requirements.
- Initially, IAM users have 0 permissions when created.
- Root user assigns permissions to IAM users based on their roles or requirements.
- IAM users can perform only the actions allowed by the policies attached to them.



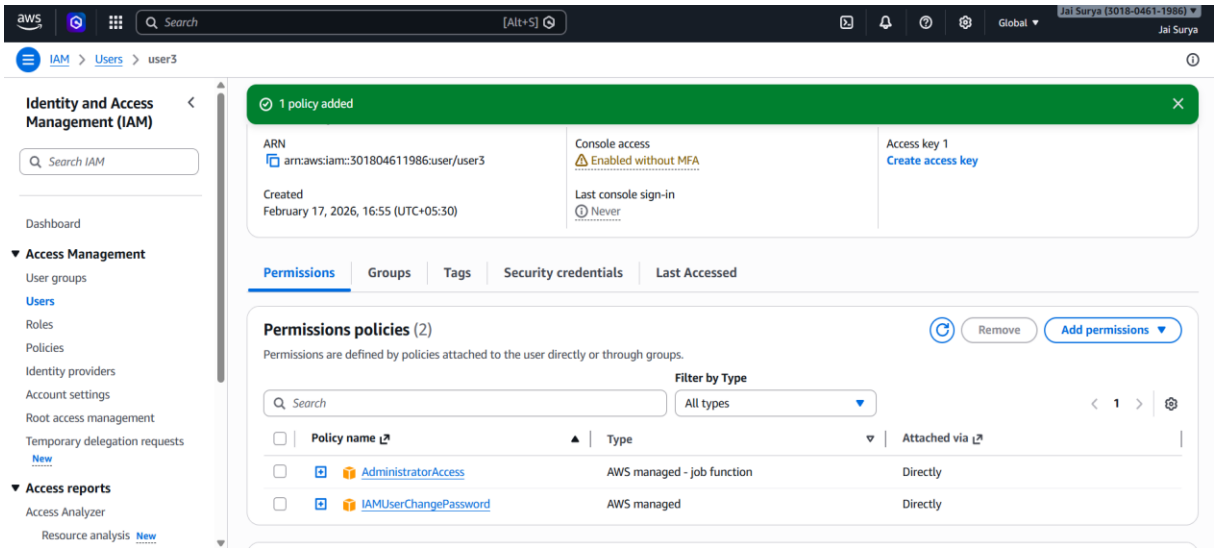
CREATING IAM USER

- Sign in as Root User or IAM Admin.
- Go to IAM and click “Add Users.”
- Enter username and select access type.
- Review details and click “Create User.”
- Save credentials for future use.



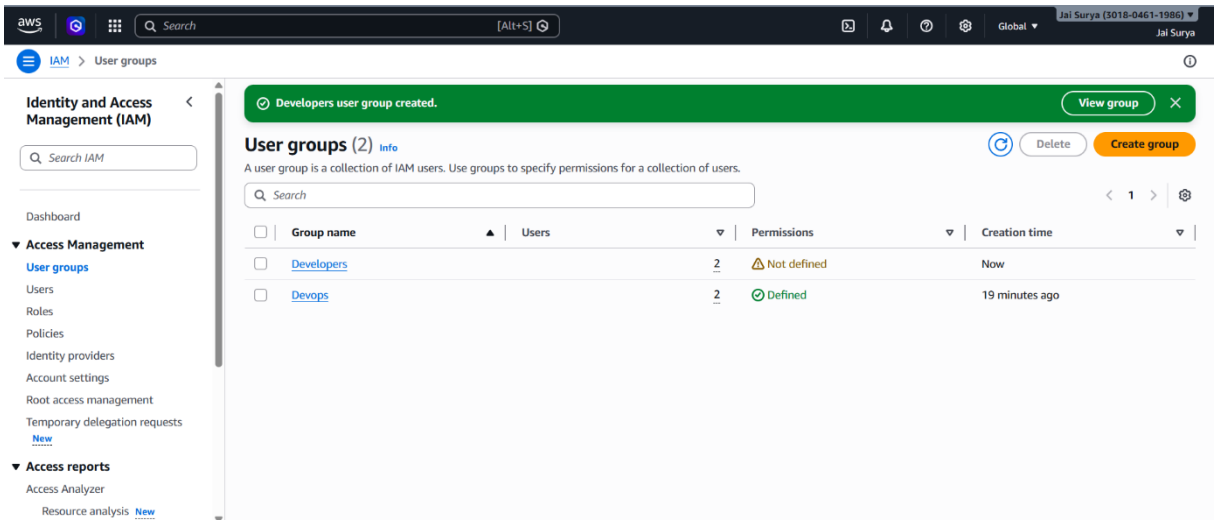
ADD PERMISSIONS TO A SINGLE IAM USER:

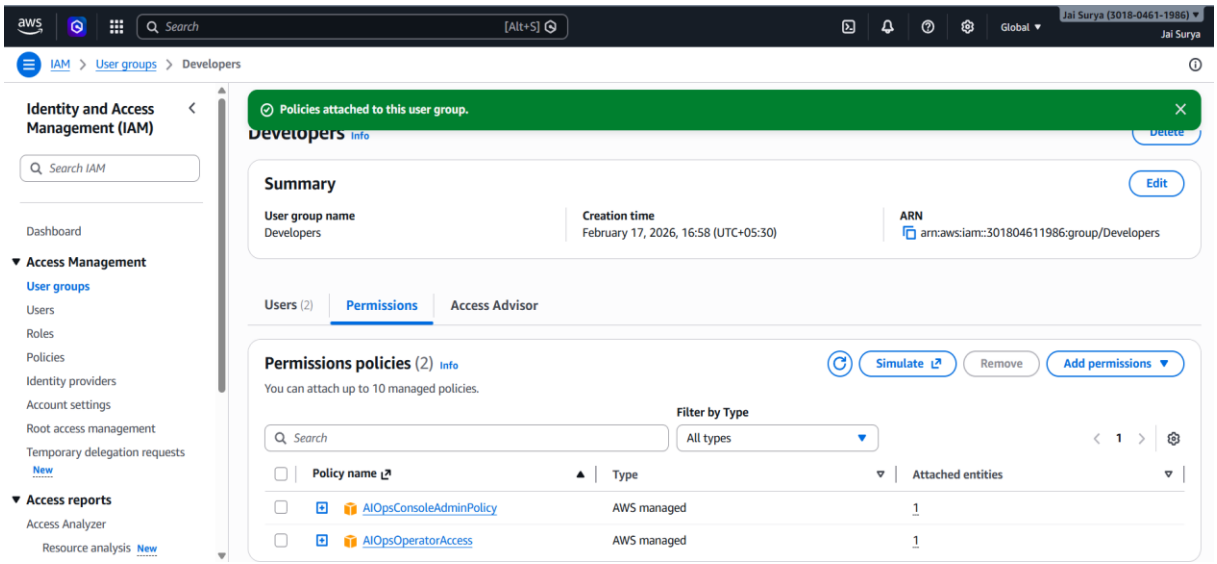
- Sign in as Root User or IAM Admin.
- Go to IAM and select the specific IAM user.
- Go to the “Permissions” tab and click “Add permissions.”
- Attach the required policy directly to the user.
- Click “Add permissions” to save.



CREATE AN IAM GROUP AND ADD USERS:

- Sign in as Root User or IAM Admin.
- Go to IAM and click “Groups.”
- Click “Create New Group” and Enter a group name.
- Enter a group name.
- Add users to the group by selecting them from the list.
- Click “Create Group” to finish.



ADD PERMISSIONS TO AN IAM GROUP:

- Sign in as Root User or IAM Admin.
- Go to IAM and click “Groups.”
- Select the specific group.
- Go to the “Permissions” tab and click “Add permissions.”
- Attach the required policies to the group and Click “Add permissions” to save.